



Evaluation of Anemia

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Anemia is a decrease in the number of red blood cells (RBCs) as measured by the red cell count, the hematocrit, or the red cell hemoglobin content.

In adult males, anemia is defined as any of the following:

- Hemoglobin < 13.6 g/dL (< 1360 g/L)
- Hematocrit < 40% (< 0.40)
- RBC < 4.5 million/mcL (< $4.5 \times 10^{12}/L$)

In adult females, anemia is defined as any of the following:

- Hemoglobin < 12 g/dL (< 120 g/L)
- Hematocrit < 37% (< 0.37)
- RBC < 4 million/mcL (< $4 \times 10^{12}/L$)

For infants and children, normal values vary with age, necessitating use of age-related tables (see table [Age-Specific Values for Hemoglobin and Hematocrit](#)).

Anemia is not a diagnosis; it is a manifestation of an underlying disorder (see [Etiology of Anemia](#)). Thus, even mild, asymptomatic anemia should be investigated so that the primary problem can be diagnosed and treated.

Anemia is usually suspected based on the history and physical examination. Common symptoms and signs of anemia include

- General fatigue
- Weakness
- Dyspnea on exertion
- Pallor

History and physical examination are followed by laboratory testing with a complete blood count, reticulocyte count, and peripheral smear. The differential diagnosis (and cause of anemia) can then be further refined based on the results of testing.

Patient History in Anemia

The history should address

- Risk factors for particular anemias
- Symptoms of anemia itself
- Symptoms that reflect the underlying disorder

Risk factors for anemia

Anemia has many risk factors. For example, a vegan diet predisposes to [vitamin B12 deficiency anemia](#), whereas alcohol use disorder increases the risk of [folate deficiency anemia](#). A number of [hemoglobinopathies](#) are inherited, and certain [medications and infections predispose to hemolysis](#). Cancer, rheumatic disorders, and chronic inflammatory disorders can suppress red cell production. Autoimmune disorders such as [systemic lupus erythematosus](#) or [lymphoma](#) can predispose to [autoimmune hemolytic anemia](#).

Pearls & Pitfalls

- Anemia is not a diagnosis; it is a manifestation of an underlying disorder. Thus, even mild, asymptomatic anemia should be investigated so that the primary problem can be diagnosed and treated.

Symptoms of anemia

The symptoms of anemia are neither sensitive nor specific and do not help differentiate between types of anemias. Symptoms reflect compensatory responses to tissue hypoxia and usually develop when the hemoglobin level falls well below the patient's individual baseline. Symptoms are generally more pronounced in patients with limited cardiopulmonary reserve or when the anemia develops very rapidly.

Symptoms such as weakness, fatigue, drowsiness, angina, syncope, and dyspnea on exertion can indicate anemia. Vertigo, headache, pulsatile tinnitus, amenorrhea, loss of libido, and gastrointestinal (GI) symptoms may also occur.

[Heart failure](#) or [shock](#) can develop in patients with severe tissue hypoxia or hypovolemia.

Symptoms that suggest the cause of anemia

Certain symptoms may suggest the cause of the anemia. For example, melena, epistaxis, hematochezia, hematemesis, or menorrhagia indicates bleeding. Jaundice and dark urine, in the absence of liver disease, suggest hemolysis. Weight loss may suggest cancer. Diffuse severe bone or chest pain may suggest [sickle cell disease](#), and stocking-glove paresthesias may suggest [vitamin B12 deficiency](#).

Physical Examination in Anemia

A complete physical examination is necessary. Signs of anemia itself are neither sensitive nor specific; however, pallor is common with severe anemia (ie, hemoglobin < 7 g/dL [< 70 g/L]).

Signs of many underlying disorders are more diagnostically accurate than are signs of anemia. Heme-positive stool identifies [gastrointestinal bleeding](#). Hemorrhagic [shock](#) (eg, hypotension, tachycardia, pallor, tachypnea, diaphoresis, confusion) may result from acute bleeding. [Jaundice](#) may suggest hemolysis. [Splenomegaly](#) may occur with hemolysis, hemoglobinopathies, systemic rheumatic diseases, myeloproliferative disorders, infections, or cancers. [Peripheral neuropathy](#) suggests [vitamin B12 deficiency](#). Fever and heart murmurs suggest [infective endocarditis](#). Rarely, high-output [heart failure](#) develops as a compensatory response to anemia-induced tissue hypoxia.

Testing in Anemia

- Complete blood count (CBC) with white blood cell (WBC) and platelet counts
- RBC indices and morphology
- Reticulocyte count
- Peripheral smear
- Other blood and urine studies to determine etiology
- Sometimes bone marrow aspiration and biopsy

Laboratory evaluation begins with a complete blood count (CBC), including white blood cell (WBC) and platelet counts, RBC indices and morphology, and examination of the peripheral smear. RBC indices include mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and red blood cell distribution width (RDW). The reticulocyte count demonstrates how well the bone marrow is compensating for the anemia.

Subsequent tests are selected on the basis of these results and on the clinical presentation. These tests aid in further characterizing the anemia and determining its cause and may include blood chemistries, ferritin, haptoglobin, direct antiglobulin test, folate, and vitamin B12 measurement, as well as urinalysis, fecal occult blood determination, and

sometimes bone marrow aspiration and biopsy. Recognition of general diagnostic patterns can expedite the diagnosis (see table [Characteristics of Common Anemias](#)).

TABLE		
Characteristics of Common Anemias		
Etiology or Type	Morphologic Changes	Special Features
Blood loss, acute	Normochromic-normocytic, with polychromatophilia	If severe, possible nucleated RBCs and left shift of the WBCs Leukocytosis Thrombocytosis
Blood loss, chronic	Same as iron deficiency	Same as iron deficiency
Folate deficiency	Same as vitamin B12 deficiency	Serum folate < 5 ng/mL (< 11 nmol/L) RBC folate < 225 ng/mL RBCs (< 510 nmol/L) Nutritional deficiency and malabsorption (in sprue, pregnancy, infancy, or alcohol use disorder)
Hereditary spherocytosis	Spheroidal microcytes Normoblastic erythroid hyperplasia Reticulocytosis	Increased MCHC Increased RBC osmotic fragility Shortened survival of labeled RBCs
Warm antibody hemolytic anemia	Spheroidal microcytes Normoblastic erythroid hyperplasia Reticulocytosis	Increased MCHC Increased RBC osmotic fragility Shortened survival of labeled RBCs Positive direct antiglobulin test
Hemolysis	Normochromic-normocytic Reticulocytosis Marrow erythroid hyperplasia	Increased serum indirect bilirubin and LDH, low haptoglobin Increased stool and urine urobilinogen Hemoglobinuria in intravascular hemolysis Hemosiderinuria
Infection, cancer, chronic inflammation, or renal disease	Normochromic-normocytic early, then microcytic Normoblastic marrow Normal or increased iron stores	Decreased serum iron Decreased total iron-binding capacity Elevated serum ferritin Normal or increased marrow iron content
Iron deficiency	Microcytic, with anisocytosis and poikilocytosis Reticulocytopenia Hyperplastic marrow, with delayed hemoglobinization	Possible achlorhydria, smooth tongue, angular cheilosis, and spoon nails Absent stainable marrow iron Low serum iron

Etiology or Type	Morphologic Changes	Special Features
		Increased total iron-binding capacity Low serum ferritin
Marrow failure	Normochromic-normocytic (may be macrocytic) Reticulocytopenia Failed marrow aspiration (often) or evident hypoplasia of erythroid series or of all elements	Idiopathic (> 50%), hereditary, or secondary to exposure to toxic drugs or chemicals (eg, chloramphenicol , quinacrine, hydantoins, insecticides) or parvovirus B19
Marrow replacement (myelophthisis)	Anisocytosis and poikilocytosis (eg, teardrop cells) Nucleated RBCs Early granulocyte precursors Marrow aspiration possibly failing or showing leukemia, myeloma, myelofibrosis, or metastatic cells	Marrow infiltration with infectious granulomas, tumors, fibrosis, or lipid histiocytosis Possible hepatomegaly and splenomegaly Possible bone changes (osteosclerosis)
Cold agglutinin disease	Macrocytosis Red cell agglutination Reticulocytosis	Follows exposure to cold Results from a cold agglutinin or hemolysin test Sometimes postinfectious (paroxysmal cold hemoglobinuria)
Paroxysmal nocturnal hemoglobinuria	Leukopenia Thrombocytopenia Reticulocytosis	Hemoglobinuria, sometimes with frankly dark morning urine Hemosiderinuria Thrombosis Smooth muscle dystonia causing abdominal pain, esophageal spasm, erectile dysfunction Variable degree of bone marrow failure (cytopenias)
Sickle cell disease	Anisocytosis and poikilocytosis Some sickle cells in peripheral smear Sickling of all RBCs in preparation with hypoxia or hyperosmolar exposure Reticulocytosis	Most common in people of African ancestry in the United States Urinary hyposthenuria Hb S detected during electrophoresis Bone changes on x-ray Stroke and other thrombotic events Possibly painful vaso-occlusive crises and leg ulcers, stroke
Sideroblastic anemia	Usually hypochromic but dimorphic with normocytes and macrocytes	Inborn or acquired metabolic defect (eg, due to substances such as alcohol, isoniazid)

Etiology or Type	Morphologic Changes	Special Features
	Hyperplastic marrow, with delayed hemoglobination Ringed sideroblasts	Usually stainable marrow iron (plentiful), but can be reduced Some congenital forms respond to vitamin B6 administration Can be part of a myelodysplastic syndrome
Thalassemia	Microcytic Target cells Basophilic stippling Anisocytosis and poikilocytosis in homozygotes Nucleated RBCs in homozygotes Reticulocytosis	Elevated Hb A2 and Hb F (in beta-thalassemia) Mediterranean ancestry (common) In homozygotes, anemia from infancy Splenomegaly Bone changes on radiographs
Vitamin B12 deficiency	Oval macrocytes Anisocytosis Reticulocytopenia Hypersegmented WBCs Megaloblastic marrow	Serum B12 < 200 pg/mL (< 145 pmol/L) Elevated methylmalonic acid and homocysteine Increased LDH Antibodies to intrinsic factor in serum (pernicious anemia) Frequently, GI and CNS symptoms
CNS = central nervous system; GI = gastrointestinal; Hb = hemoglobin; LDH = lactic dehydrogenase; MCHC = mean corpuscular hemoglobin concentration; RBC = red blood cell; WBC = white blood cell.		

Complete blood count and RBC indices

The automated CBC directly measures hemoglobin, RBC count, WBC count, and platelet count, plus mean corpuscular volume (MCV), which is a measure of RBC volume. Hematocrit, which is a measure of the percentage of blood made up of RBCs, mean corpuscular hemoglobin (MCH), which is a measure of the hemoglobin content in individual RBCs but is of no clinical significance, and mean corpuscular hemoglobin concentration (MCHC), which is a measure of the hemoglobin concentration in individual RBCs are calculated values.

The diagnostic criterion for anemia is

- For adult males: Hemoglobin < 13.6 g/dL (< 136 g/L), hematocrit < 40% (< 0.40), or RBC < 4.5 million/mcL (< 4.5 × 10¹²/L)
- For adult females: Hemoglobin < 12 g/dL (< 120 g/L), hematocrit < 37% (< 0.37), or RBC < 4 million/mcL (< 4 × 10¹²/L)

For infants and children, normal values vary with age, necessitating use of age-related tables (see table [Age-Specific Values for Hemoglobin and Hematocrit](#)).

RBC populations are termed microcytic (small cells) if MCV is < 80 fL, and macrocytic (large cells) if MCV is > 100 fL. However, because reticulocytes are also larger than mature red cells, large numbers of reticulocytes can elevate the MCV.

Automated techniques can also determine the degree of variation in RBC size, expressed as the RBC volume distribution width (RDW). A high RDW may be the only indication of simultaneous microcytic and macrocytic disorders; such a pattern may result in a normal MCV, which measures only the mean value. The term hypochromia refers to RBC populations in which the MCHC is < 30%. RBC populations with a normal MCHC value are normochromic. Spherocytes can have an elevated MCHC. Red cells defend their MCHC

over their MCV (preserving hemoglobin at the expense of RBC size), which is why microcytosis occurs with iron deficiency, impaired hemoglobin synthesis, or the erythrocytosis in [polycythemia vera](#).

The RBC indices can help indicate the mechanism of anemia and narrow the number of possible causes.

Microcytic indices occur with altered heme or globin synthesis. The most common causes are [iron deficiency](#), [thalassemia](#), and related hemoglobin-synthesis defects. In some patients with the [anemia of chronic disease](#), the MCV is microcytic or borderline microcytic.

Macrocytic indices occur with impaired DNA synthesis (eg, due to [vitamin B12 deficiency](#), [folate deficiency](#), or chemotherapeutic drugs such as [hydroxyurea](#) or antifolate agents) and in [alcohol use disorder](#) because of abnormalities of the cell membrane. Acute bleeding may briefly produce macrocytic indices because of the early release of large young reticulocytes.

Normocytic indices occur in anemias resulting from deficient erythropoietin (EPO) production or inadequate response to it (hypoproliferative anemias). Hemorrhage, before iron deficiency develops, usually results in normocytic and normochromic anemia unless the number of reticulocytes is excessive.

Peripheral smear

The peripheral smear can suggest the presence or cause of a hemolytic anemia, demonstrate alterations in RBC structure, confirm the presence of thrombocytopenia (versus artifactual platelet clumping) and be used to identify immature granulocyte and abnormal lymphocyte morphology, often flagged by automated blood cell analyzers. It can detect other abnormalities (eg, malaria and other parasites, intracellular RBC or granulocyte inclusions) that can occur despite normal automated blood cell counts. RBC injury may be identified by finding RBC fragments (schistocytes), oxidative damage (bite or blister cells) or evidence of significant membrane alterations from sickled cells, oval-shaped cells (ovalocytes) or spherocytic cells. Target cells (thin RBCs with a central dot of hemoglobin) are RBCs with insufficient hemoglobin or excess cell membrane (eg, due to hemoglobinopathies or liver disorders). The peripheral smear can also reveal variation in RBC shape (poikilocytosis) and size (anisocytosis).

Peripheral Blood Smear

Peripheral Blood Smear, Normal

A drop of blood is applied against a glass slide that is subsequently stained with polychrome stains (Wright-Giemsa) to permit identification of the various cell types. These stains are mixtures of basic dyes (methylene blue) that stain as blue and acidic dyes (eosin) that stain as red. Thus, acid components of the cell (nucleus, cytoplasmic RNA, basophilic granules) stain blue or purple, and basic components of the cell (hemoglobin, eosinophilic granules) stain red or orange.

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Schistocytes (Red Blood Cell Fragments)

Schistocytes (see arrows) are damaged red blood cells, which may occur in microangiopathic hemolytic anemia (including disseminated intravascular coagulation, thrombotic thrombocytopenic purpura/hemolytic-uremic syndrome, and valvular hemolysis).

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Target Cells from Thalassemia Patient

Target cells (thin RBCs with a central dot of hemoglobin; arrow) are due to an imbalance between the volume of the cell and its hemoglobin content and characterize thalassemia, other hemoglobinopathies (eg, hemoglobin C disease and hemoglobin S-C disease) but may also occur after splenectomy and in liver disease.

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Reticulocyte count

The reticulocyte count is expressed as the percentage of reticulocytes or as the absolute reticulocyte count. The reticulocyte count is a crucial test in the evaluation of anemia because it informs about the response of the bone marrow and facilitates differentiation between deficient erythropoiesis (RBC production) and excessive hemolysis (RBC destruction) as the cause of anemia. For example, higher values indicate compensatory production (reticulocytosis); in the presence of anemia, reticulocytosis suggests excessive RBC destruction, blood loss, or recovery from anemia (e.g., following iron repletion for iron deficiency anemia). Low or normal range numbers in the presence of anemia indicate inappropriately low RBC production. However, in some cases of hemolytic anemia the reticulocyte count will be low (e.g. comorbid renal disease, infection) and these patients are at risk for developing severe anemia. Calculating the reticulocyte production index corrects the reticulocyte count for anemia.

Reticulocytes are best visualized when blood is stained with a supravital stain. Red cell reticulin is composed of RNA, which is present only in young red cells, so reticulocytes will only have a bluish appearance in a Wright-stained blood smear (polychromatophilia or polychromasia). The presence of polychromatophilia can provide a rough estimate of reticulocyte production on a routine blood smear.

Bone marrow aspiration and biopsy

Bone marrow aspiration and biopsy provide direct observation and assessment of RBC precursors. The presence of abnormal maturation (dyspoiesis) of blood cells and the amount, distribution, and cellular pattern of iron content can be assessed. Bone marrow aspiration and biopsy are usually not indicated in the evaluation of anemia and are only

done when one of the following conditions is present:

- Unexplained anemia
- More than one cell lineage abnormality (ie, concurrent anemia and thrombocytopenia or leukopenia)
- Suspected primary bone marrow disorder (eg, [leukemia](#), [multiple myeloma](#), [aplastic anemia](#), [myelodysplastic syndrome](#), metastatic carcinoma, [myelofibrosis](#))

Cytogenetic and molecular analyses can be done on aspirate material in hematopoietic or other tumors or in suspected congenital lesions of RBC precursors (eg, Fanconi anemia). Flow cytometry can be done in suspected lymphoproliferative or myelodysplastic states to define the immunophenotype.

Bone marrow aspiration and biopsy are not technically difficult and do not pose significant risk of morbidity. These procedures are safe and helpful when hematologic disease is suspected. Bone marrow aspiration and biopsy are usually done as a single procedure. Because biopsy requires adequate bone depth, the sample is usually taken from the posterior (or, less commonly, anterior) iliac crest. If myeloma is suspected, there is severe osteoporosis, or body habitus presents procedural challenges, imaging-guided biopsy can be done.

Other tests for evaluation of anemia

Serum indirect bilirubin and lactate dehydrogenase (LDH) can sometimes help differentiate between hemolysis and blood loss. Both are elevated in hemolysis and normal in blood loss, although reabsorption of a large hematoma can decrease haptoglobin and elevate indirect bilirubin, thus mimicking a hemolytic anemia. However, LDH and bilirubin can be elevated for reasons other than hemolytic anemia and are not specific.

Other tests, such as vitamin B12 and folate levels and iron and iron binding capacity, are done depending on the suspected cause of anemia. Other tests are discussed under specific anemias and bleeding disorders.

Drug Information for the Topic

CLINICAL CALCULATORS

[Absolute Reticulocyte Count](#)

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CLINICAL CALCULATORS

[Reticulocyte Production Index](#)

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Bone Marrow Biopsy

VIDEO

Bone Marrow Biopsy & Aspiration

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